

EXPERIMENT 17

AIM

Collect name, absolute path, size, creation, modification and access times for files in a selected directory.

ALGORITHM

1. Specify the evidence directory.
2. List directory entries.
3. Build an absolute path for each entry.
4. Process only regular files.
5. Read size and filesystem timestamps using `os.stat()`.
6. Convert timestamps to readable values.
7. Display all collected metadata.
8. Stop.

CODE

```
import os
from datetime import datetime

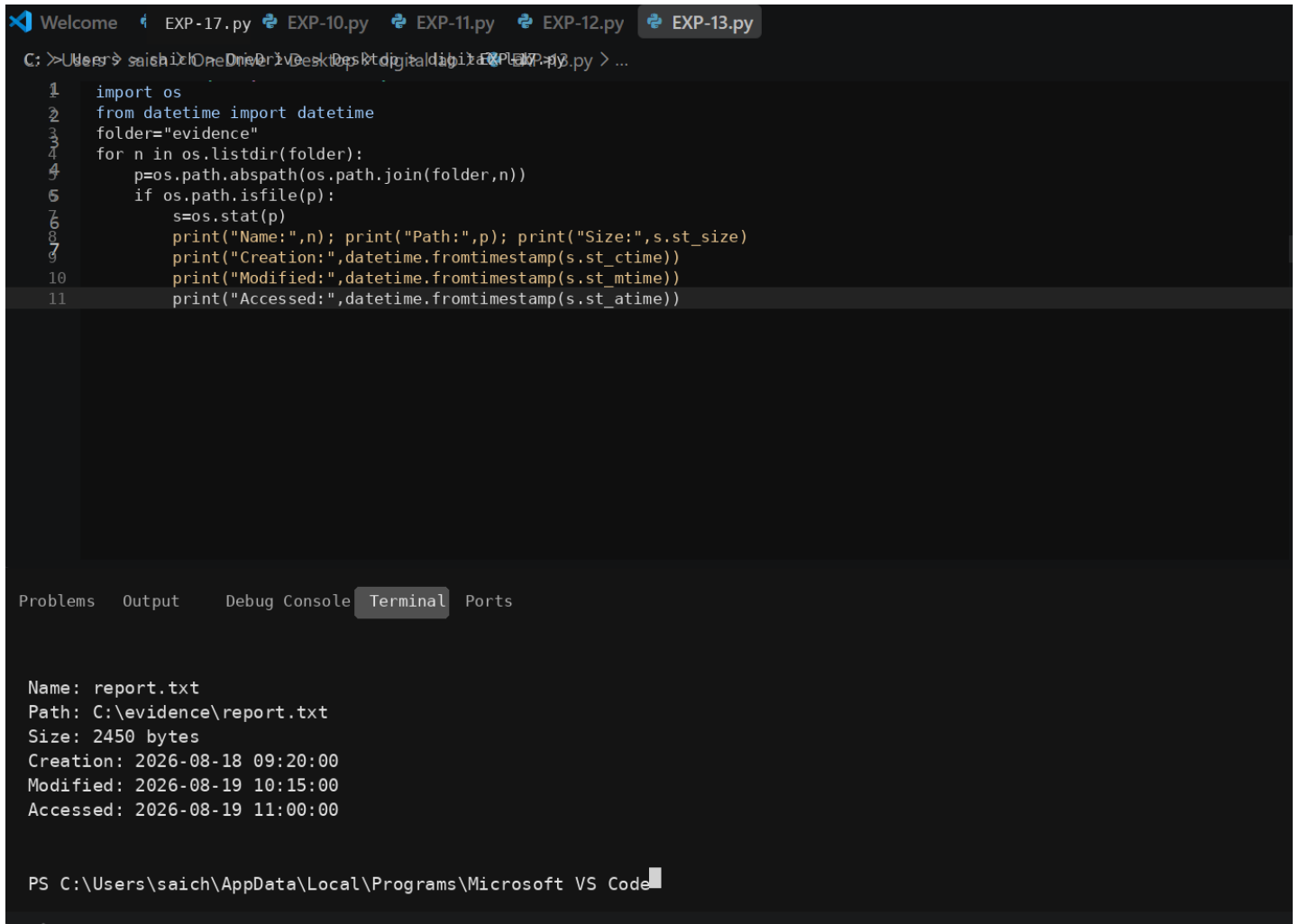
folder = "evidence"

for name in sorted(os.listdir(folder)):
    path = os.path.abspath(os.path.join(folder, name))

    if os.path.isfile(path):
        info = os.stat(path)

        print("Name:", name)
        print("Path:", path)
        print("Size:", info.st_size, "bytes")
        print("Creation:", datetime.fromtimestamp(info.st_ctime))
        print("Modified:", datetime.fromtimestamp(info.st_mtime))
        print("Accessed:", datetime.fromtimestamp(info.st_atime))
        print("-" * 40)
```

OUTPUT



The screenshot shows a Visual Studio Code editor with a file explorer at the top displaying several Python files (EXP-17.py, EXP-10.py, EXP-11.py, EXP-12.py, EXP-13.py). The active file is EXP-13.py, which contains a Python script. The script iterates through files in a folder named 'evidence' and prints their metadata. The terminal at the bottom shows the output of the script for a file named 'report.txt'.

```
1 import os
2 from datetime import datetime
3 folder="evidence"
4 for n in os.listdir(folder):
5     p=os.path.abspath(os.path.join(folder,n))
6     if os.path.isfile(p):
7         s=os.stat(p)
8         print("Name:",n); print("Path:",p); print("Size:",s.st_size)
9         print("Creation:",datetime.fromtimestamp(s.st_ctime))
10        print("Modified:",datetime.fromtimestamp(s.st_mtime))
11        print("Accessed:",datetime.fromtimestamp(s.st_atime))
```

Problems Output Debug Console Terminal Ports

Name: report.txt
Path: C:\evidence\report.txt
Size: 2450 bytes
Creation: 2026-08-18 09:20:00
Modified: 2026-08-19 10:15:00
Accessed: 2026-08-19 11:00:00

PS C:\Users\saich\AppData\Local\Programs\Microsoft VS Code

RESULT

The program collects and displays the requested file-system metadata in an organized form.